

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA
Faculty of Engineering, Built Environment and Information Technology

School of Information Technology

Department of Computer Science

COS326 – Database Systems

Practical 5 2026

Release Date:	20 October 2026
Submission Date and Time:	01 November 2026 @ 23:59Hrs
Lecturer:	Mr S.M Makura
Total: 50 Marks	

A. Objectives

1. Import structured CSV data into a Neo4j graph database.
2. Create nodes and relationships using Cypher.
3. Retrieve, filter, sort and aggregate graph data.
4. Execute path queries and identify shortest paths in a graph.
5. Work with multiple node labels and relationship types.
6. Develop a Java application that connects to Neo4j using the Neo4j Java Driver.
7. Execute database transactions and display query results through a graphical user interface.
8. Apply graph database techniques to analyse a social network.

B. Submission Procedure

Submit a single compressed archive containing:

1. Task1Queries.txt — all Cypher queries for Tasks 1–3.
2. The complete Java source code and any necessary project configuration files.
3. A README.txt file containing instructions for configuring the database connection and running the application.
4. Screenshots demonstrating the successful execution of the import queries, representative graph queries and the Java application's main functions.

Name your archive using your student number, for example:
`u12345678-prac5.zip`
Upload the archive to ClickUP before the submission deadline.

NO LATE submissions will be accepted after the submission date and time has lapsed. Do not wait until the last minute to submit and then provide reasons related to technical difficulties.

Academic Integrity: You may use AI-assisted tools (e.g., ChatGPT, Copilot, Gemini etc) to help you understand Neo4J concepts and syntax. However, you remain responsible for the design, implementation, and correctness of your solution. During marking, you may be asked to explain or modify any part of your submission. Failure to demonstrate understanding may result in marks being deducted.

Question 1: Social Network Investigation

Scenario

A cybersecurity investigation team is analysing a social networking platform following reports that suspicious information is being circulated through a network of accounts.

The investigation team has received two datasets:

- **persons.csv** - contains account identifiers and account names.
- **follows.csv** - contains directed relationships indicating which accounts follow other accounts.

The investigation team wants to understand how accounts are connected, identify influential accounts, investigate connections between accounts, and analyse how information can spread through the network.

The team also requires a small desktop application that allows investigators to query the Neo4j database without manually executing Cypher statements.

You have been appointed as a junior database developer to implement the required graph database and application.

Important: The supplied datasets describe following relationships. They do not establish that any account is malicious. Any suspicious activity investigated in this practical must be treated as a hypothetical scenario rather than a conclusion about the supplied accounts.

Source: Makura S.M (2026)

TASK 1: IMPORTING DATA AND CREATING THE GRAPH [10 Marks]

Question 1.1: Database preparation and data import [6 marks]

Create a database named COS326Prac5 and import the supplied CSV data.

a. Write a Cypher query using LOAD CSV WITH HEADERS to create Person nodes from persons.csv. Each node must contain an id and a name property. Ensure that the imported ID is stored consistently as an integer or another appropriate identifier type. (3)

b. Write a Cypher query to create FOLLOWS relationships using follows.csv. Each relationship must connect the appropriate source and destination Person nodes based on their account names. (3)

Note: Inspect the CSV files before importing them. Ensure that the import path is correct and that the property names used in your queries match the actual column headers.

Question 1.2: Verify the graph [4 marks]

a. Write a Cypher query to display the total number of Person nodes and the total number of FOLLOWS relationships. (2)

b. Write a Cypher query to display a sample of five people and their outgoing FOLLOWS relationships. (2)

Save all queries in Task1Queries.txt.

TASK 2: CYPHER QUERIES AND GRAPH OPERATIONS [12 Marks]

The investigation team needs to explore the social network and understand the structure of the graph.

Write and execute a Cypher query for each of the following requirements.

Question 2.1: Basic graph queries [4 marks]

a. List all distinct node labels present in the database. (1)

b. Retrieve the names of all people in alphabetical order. Display the first 20 results only. (1)

c. List the distinct relationship types present in the graph. (1)

d. Retrieve the names of people who follow a specified account. Your query must use a parameter or a clearly identified account name. (1)

Question 2.2: Filtering and aggregation [4 marks]

a. Identify the five accounts with the highest number of incoming FOLLOWS relationships. Display the account name and follower count in descending order. (2)

b. For each person, calculate the number of accounts they follow. Display the five people with the highest outgoing following counts. (2)

Question 2.3: Creating and querying content [4 marks]

The investigation team has requested that the graph also represent information shared on the platform.

a. Create at least three Post nodes. Each post must have a unique identifier, a hashtag and a message. Use the following illustrative data:

Post ID	Hashtag	Message
P001	#Security	Verify suspicious links
P002	#Awareness	Protect your account
P003	#Updates	Review account activity

Create a POSTED relationship between an existing Person node and each post. (2)

b. Create at least one REPOSTED relationship and one LIKED relationship between appropriate Person and Post nodes. Each relationship must connect existing nodes and must not create duplicate copies of the nodes. (1)

c. Write a query to display the account names, hashtags and messages of posts that have been reposted. (1)

Save all queries in Task1Queries.txt.

TASK 3: PATH ANALYSIS AND NETWORK INVESTIGATION [12 Marks]

The investigation team wants to understand how accounts are connected and how information could travel through the network.

Question 3.1: Path queries [6 marks]

a. Given two account names, write a Cypher query to find accounts reachable from the first account by following one or two FOLLOWS relationships. Display the reachable account names and the path lengths. Avoid returning the starting account as its own destination. (2)

b. Given two account names, write a Cypher query to display the nodes and relationships along a path between them. The query must handle the possibility that no path exists. (2)

c. Write a Cypher query to find the shortest path between two specified accounts using only FOLLOWS relationships. The query must handle the case where no path is found. (2)

For questions (a)-(c), use parameterised account names or clearly indicate where the investigator should enter the account names. Remember that FOLLOWS is a directed relationship.

Question 3.2: Investigation reports [6 marks]

a. Write a query to identify people who have posted content and indicate, for each person, whether at least one of their posts has been reposted. Display the person's name and a Boolean result. (2)

b. Write a query to calculate the total number of posts, the number of reposted posts and the number of liked posts. Avoid counting the same post multiple times when it has several relationships. (2)

c. Write a query to identify the accounts that are directly connected to a specified account through either an incoming or outgoing FOLLOWS relationship. Display the account names and the direction of the relationship. (2)

Save all queries in Task1Queries.txt.

TASK 4: JAVA APPLICATION FOR SOCIAL NETWORK ANALYSIS [16 Marks]

Develop a Java desktop application using the Neo4j Java Driver. The application must connect to the COS326Prac5 database and allow an investigator to analyse the social network through a graphical interface.

You may use JavaFX or Swing.

Question 4.1: Database connectivity [3 marks]

a. Establish a connection to the Neo4j database using the Neo4j Java Driver. (2)

b. Display an appropriate message indicating whether the connection was successful or failed. (1)

Question 4.2: Graphical user interface [3 marks]

Develop an appropriate graphical interface that includes:

- A field for entering an account name where required.
- Controls for selecting or executing the required analyses.
- A component for displaying query results.

- Appropriate error messages and basic input validation.

(3)

Question 4.3: Implement the investigation functions [8 marks]

The application must provide the following functions.

a. **Network overview:** Display the total number of people and the total number of FOLLOWS relationships. (2)

b. **Influential accounts:** Display the five accounts with the highest number of followers, including the follower counts. (2)

c. **Connection investigation:** Allow the investigator to enter two account names and display the shortest available path between them using FOLLOWS relationships. If no path exists, display an appropriate message. (2)

d. **Content investigation:** Display the posts associated with a specified account, including the hashtag, message and whether each post has been reposted. (2)

Question 4.4: Transactions and error handling [2 marks]

a. Use an appropriate transaction mechanism when executing database operations. (1)

b. Handle database errors and invalid account names without terminating the application unexpectedly. (1)

Implementation requirements:

- Do not hard-code the query results in the application.
- Execute the relevant Cypher queries against Neo4j.
- Use appropriate parameters when supplying account names or other input values.
- Display actual query results in the graphical interface.
- Close database sessions and the driver appropriately when they are no longer required.